Mask-Aware Temporal Keypoint Imputation for Robust Low-Dimensional Diffusion Policy Push-T

Final Project Report
Robotics Final Project

Abstract: This project reproduces the official low-dimensional Push-T checkpoint from Diffusion Policy and studies a deployment failure mode that is not handled by the original low-dimensional policy interface. The Push-T keypoint runner exposes both keypoint coordinates and a visibility mask, but the low-dimensional policy only consumes the coordinate vector. In a realistic detector-style deployment, invisible keypoints may be missing or zero-filled before policy inference, creating out-of-distribution observations for a checkpoint trained on fully visible keypoints. We implement a training-free, mask-aware observation wrapper that imputes masked keypoints before normalization and policy inference. On the official checkpoint with 20 matched Push-T seeds, per-keypoint temporal imputation substantially improves robustness under synthetic keypoint dropout. At 50% keypoint visibility, zero-fill obtains score 0.177 and no successes, while carry-forward reaches 0.901 score and 0.75 success and linear extrapolation reaches 0.875 score and 0.85 success. At 25% visibility, linear extrapolation improves score from 0.134 to 0.668. The contribution is intentionally bounded: this is a low-dimensional Push-T deployment wrapper, not a general visual Diffusion Policy algorithm.

Keywords: Diffusion Policy, Push-T, keypoint occlusion, imputation, robot learning

1 Introduction

Diffusion Policy [1] models robot actions as denoised action chunks and has become a common baseline for visuomotor imitation learning. The official Push-T low-dimensional setting uses nine block keypoints and the agent position as observations. The official keypoint environment also produces a keypoint visibility mask, but the low-dimensional policy interface normalizes and consumes only `obs_dict["obs"]`. Thus, the default deployment path discards visibility information before the policy can use it.

This project started from the official Push-T checkpoint reproduction and searched for an improvement that could be validated directly on the official checkpoint. Earlier scheduler and DDIM directions were downgraded after review: the scheduler result relied on a hand-designed surrogate, and DDIM is already an established sampler. The final contribution is instead a narrow deployment robustness improvement:

> We identify a mask-handling failure mode in the official low-dimensional Push-T Diffusion Policy deployment path and show that a training-free, mask-aware temporal observation wrapper substantially improves robustness under detector-style keypoint dropout.

9th Conference on Robot Learning (CoRL 2025), Seoul, Korea.

2 Source Audit and Problem Setting

The source audit is central to the claim boundary. In the official low-dimensional policy, `predict_action` normalizes only `obs_dict["obs"]`. The Push-T keypoint runner constructs both fields:

```
'obs': obs[...,:self.n_obs_steps,:Do]
'obs_mask': obs[...,:self.n_obs_steps,Do:] > 0.5
```

but the policy never reads `obs_mask`. The environment samples keypoint visibility using `keypoint_visible_rate`; however, its low-dimensional observation still stores the true keypoint coordinates, while the zeroed visible-keypoint copy is used for rendering. Therefore, to evaluate detector-style occlusion, we explicitly zero masked keypoint coordinates before policy inference and then compare imputation strategies.

The training pipeline further supports this interpretation. The low-dimensional dataset reads `keypoint`, `state`, and `action` from the replay buffer and concatenates keypoints with agent position. It has no mask channel and no keypoint dropout augmentation. The official training configuration sets `keypoint_visible_rate: 1.0`. Thus, the checkpoint is trained on full keypoint observations; zero-filled detector dropout is an out-of-distribution deployment input, not a condition the policy was trained to handle.

3 Method

We implement a lightweight `DeploymentWrapper` around the loaded official policy. The wrapper changes only the observation passed into `predict_action`; it does not change the checkpoint weights, normalizer, sampler, action horizon, or environment dynamics. Imputation is performed in the original Push-T coordinate space before the policy normalizer.

Let $x_{t,j}$ be a keypoint coordinate and $m_{t,j}$ its visibility mask. If $m_{t,j} = 1$, the true coordinate is used. If $m_{t,j} = 0$, we evaluate:

- **zero-fill**: pass zeroed masked coordinates directly;
- **mean prior**: fill with the checkpoint normalizer's training mean for that coordinate;
- **frame hold**: use the most recent fully visible keypoint frame, otherwise the mean prior;
- **carry-forward**: for each coordinate independently, use the most recent visible value, otherwise the mean prior;
- **linear**: for each coordinate independently, extrapolate from the two most recent visible values, clamp to $[0, 512]$, otherwise fall back to carry-forward or the mean prior;
- **oracle**: retain true coordinates despite the mask; this is an upper bound, not deployable.

The per-coordinate history is reset at episode start. If a coordinate is invisible before any observation has been seen, the fallback is the training mean, not zero.

4 Experimental Setup

Experiments use the official low-dimensional Push-T checkpoint, the official keypoint runner, DDIM inference with $(k, h) = (100, 8)$, 20 matched test seeds, and two synthetic visibility rates: 50% and 25%. The visibility masks are produced by the seeded environment and the same seed set is used for every imputation method. We report mean Push-T score, success rate at score ≥ 0.95, and paired bootstrap confidence intervals over matched seeds.

The current evaluation uses the official iid per-keypoint visibility mask. This is a favorable setting for temporal imputation because a missing keypoint is often visible again soon. Continuous occlusion is left as a limitation and future experiment.

Table 1: Official checkpoint results under detector-style masked keypoint dropout. Each entry reports mean score and success rate over 20 matched seeds.

Visibility	Method	Score	Success	Δ vs. zero
50%	zero-fill	0.177	0.00	0.000
50%	mean prior	0.405	0.25	+0.228
50%	frame hold	0.403	0.20	+0.226
50%	carry-forward	0.901	0.75	+0.725
50%	linear	0.875	0.85	+0.698
50%	oracle	0.900	0.90	+0.723
25%	zero-fill	0.134	0.00	0.000
25%	mean prior	0.257	0.00	+0.123
25%	frame hold	0.257	0.00	+0.123
25%	carry-forward	0.575	0.20	+0.440
25%	linear	0.668	0.35	+0.533
25%	oracle	0.900	0.90	+0.766

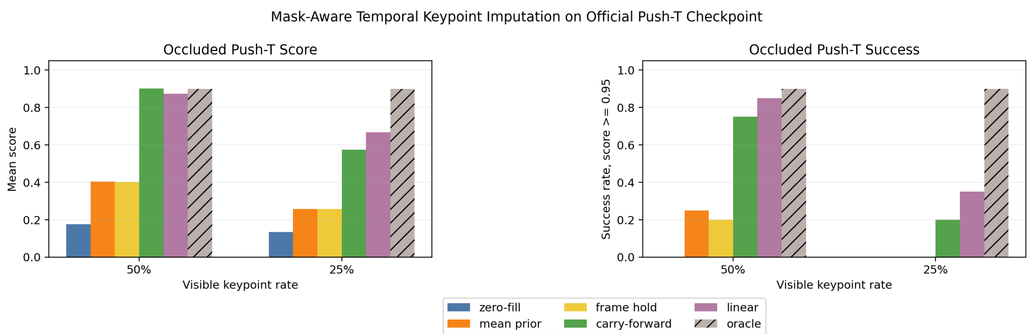

Figure 1: Mask-aware temporal imputation substantially improves official Push-T checkpoint robustness under masked keypoint dropout. Oracle is an upper bound and not deployable.

5 Results

Paired bootstrap intervals confirm that temporal imputation improves over zero-fill. At 50% visibility, carry-forward improves score by +0.725 with 95% CI $[+0.586, +0.846]$, and linear improves score by +0.698 with CI $[+0.544, +0.835]$. At 25% visibility, carry-forward improves by +0.440 with CI $[+0.290, +0.587]$, and linear improves by +0.533 with CI $[+0.378, +0.685]$.

Linear extrapolation has higher mean score and success than carry-forward at 25% visibility, but the paired difference is not statistically conclusive in this 20-seed evaluation: score delta is +0.093 with CI $[-0.125, +0.305]$, and success delta is +0.15 with CI $[-0.10, +0.40]$. Therefore, we report it as a trend rather than a strict dominance claim.

6 Discussion

The result is not merely "do not feed zeros." Mean-prior and full-frame-hold baselines improve over zero-fill but remain far below per-keypoint temporal methods. This suggests that preserving recent per-keypoint geometry is the useful signal. At 50% visibility, temporal imputation nearly reaches the oracle upper bound. At 25% visibility, it recovers a large fraction of performance but remains below oracle, so severe occlusion is not solved.

The most defensible contribution is the combination of source-level failure identification and official-checkpoint evaluation. The project does not claim to invent imputation, improve clean full-observation performance, or solve visual occlusion. It shows that the official low-dimensional deployment path discards mask information and that a simple no-retraining wrapper can make this setting much more robust.

7 Limitations

First, the occlusion model is iid per-keypoint dropout from the official environment mask. Real keypoint detectors often fail with temporally correlated occlusion, so continuous dropout should be evaluated before claiming real-world robustness. Second, the experiment is specific to low-dimensional Push-T keypoints and does not directly transfer to RGB Diffusion Policy. Third, oracle performance shows that 25% visibility still has substantial headroom. Fourth, the official reproduction runs in a compatibility environment rather than the exact original training stack.

8 Conclusion

This project reproduces the official Diffusion Policy Push-T checkpoint and implements a focused inference-time improvement. The final contribution is a training-free, mask-aware temporal observation wrapper for low-dimensional keypoint dropout. On official checkpoint rollouts with matched seeds, per-keypoint temporal imputation substantially improves robustness over zero-fill, mean-prior fill, and full-frame hold baselines. The claim is narrow but defensible and directly supported by official-checkpoint experiments.

References

[1] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. Burchfiel, and S. Song. Diffusion policy: Visuomotor policy learning via action diffusion. In *Robotics: Science and Systems*, 2023.